Lab 2 – AI^2 Product Specification

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April 4, 2026

Version 2

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a comprehensive outline of AI^2's design requirements that have been previously agreed upon. The SRS will make it clear what the software's intended behavior shall be, features it will include, as well as what it will not do, and how it will meet the needs of the intended users. These will be unambiguous guidelines for developers of the application to adhere to. This SRS will follow the IEEE 830-1998 standard.

1.2 Scope

AI^2's primary function is to match invoices from various formats received via email to a corresponding purchase order (PO) before forwarding to the accounts payable (AP) team. AI^2 will utilize both an Artificial Intelligence (AI) model and hardcoded rules to perform matching. AI^2 will support the AP team members by reducing the amount of time that will need to be spent finding matching POs for incoming invoices.

The system shall provide dashboards for AP team members, administrators, and the model trainer. These dashboards shall provide access to each user based on their user role and allow them to interact with the system, and to view the matched invoices. The administrators shall have the ability to configure the system to adhere to their company's preferences from their dashboard, as well as monitor system health and invoice matches. The AP team members shall be able to interact with the invoice matches through their dashboard. The model trainer shall be able to adjust parameters for the AI model through their dashboard. The system will not span outside of this and will not support other types of accounting documentation.

1.3 Definitions, Acronyms, and Abbreviations

AI² (Artificially Intelligent Invoices): The proposed automated invoice processing system designed to streamline Accounts Payable workflows through AI-driven matching, classification, and routing.

Alteryx: A data automation and analytics tool that enables secure API connections and data workflows between applications.

AP (Accounts Payable): The amount a business owes for goods and services purchased on credit; typically, due at intervals of 30, 45, 60, or 90 days depending on repayment terms.

APEX: A strongly typed, object-oriented programming language that Salesforce developers use to execute flow and transaction control statements on the Salesforce platform.

AWS GovCloud: A secure U.S. government-compliant cloud environment that supports sensitive data and regulated workloads in accordance with FedRAMP and DoD requirements.

Atlantic Diving Supply (ADS): An American federal contractor company that provides equipment and logistics solutions to the Department of Defense, federal agencies, and first responders.

CMMC (Cybersecurity Maturity Model Certification): A framework developed by the U.S. Department of Defense to assess and enhance the cybersecurity posture of contractors handling federal information.

DoD (Department of Defense): A department in the United States of America's Federal Executive Branch that is responsible for handling the United States' military forces.

ERP (Enterprise Resource Planning System): An integrated software system used to manage core business processes, such as finance, procurement, and supply chain operations.

FedRAMP (Federal Risk and Authorization Management Program): A U.S. government program that standardizes security assessment, authorization, and continuous monitoring for cloud services.

Graph API: A Microsoft API interface used to access and automate data operations within Outlook and other Microsoft 365 services.

Invoice: A document listing goods or services provided, including a statement of the amount due; a bill.

Oracle: The system of record used by ADS for enterprise data and financial management.

ORM (Object Relational Mapping): A technique that allows database interactions without using raw SQL.

Purchase Order (PO): The official order list ADS sends to suppliers detailing the items or services requested.

Salesforce: A robust Customer Relationship Management (CRM) platform that allows businesses to manage customer relationships efficiently.

Supplier: A person or organization that provides goods or services needed by another organization.

Vendor: A person or company offering goods or services for sale.

1.4 References

[1] "Accounts Payable Automation Trends: Key Insights and Latest Statistics." Kefron. https://kefron.com/2025/02/accounts-payableautomation-trends-key-insights-and-latest-statistics/?switch=us (accessed March 21, 2026).

[2] A. Hayes. "Understanding Accounts Payable (AP) With Examples and How to Record

AP." Investopedia. https://www.investopedia.com/terms/a/accountspayable.asp (accessed March 21, 2026).

[3] "AI Invoice Processing Benchmarks 2025 - Accuracy, Speed, And Cost Comparison." Parseur. https://parseur.com/blog/ai-invoiceprocessingbenchmarks#:~:text=Speed%20(according%20to%20SuperAGI),automation%20drives%20significant%20productivity%20gains (accessed March 21, 2026).

[4] "Apex in Salesforce - SalesForce FAQs." SalesforceFAQs.com. https://salesforcefaqs.com/apex/ (accessed March 21, 2026).

[5] "Gartner Survey Shows That a Third of Accountants Make Several Financial Errors Per Week." Gartner. https://www.gartner.com/en/newsroom/press-releases/2024-02-21-gartner-survey-showsthat-a-third-of-accountants-make-several-error-per-week-due-to-capacity-constraints (accessed March 21, 2026).

[6] "Military Equipment & Tactical Gear Suppliers." ADS Inc. https://www.adsinc.com (accessed March 21, 2026).

[7] R. Tamaro. "The Hidden Costs of Manual Accounts Payable and How Automation Solves Them - CASO Document Management." CASO Document Management. https://caso.com/2025/03/the-hidden-costs-of-manual-accounts-payable-and-howautomation-solves-them/ (accessed March 21, 2026).

[8] S. Jex. "Accounts Payable Automation by the Numbers: 10 Statistics to Know." Quadient. https://www.quadient.com/en/blog/ap-automation-by-the-numbers-10-statistics-to-know (accessed March 21, 2026).

[9] S. Todd, "Lab 1 – AI^2 Product Description," *CS 411W, Old Dominion University,* 2026.

1.5 Overview

The remaining sections of this SRS will expound upon the system architecture, product functions, user characteristics, constraints, and assumptions and dependencies of AI^2. It shall include two visual aids, the Major Functional Component Diagram (MFCD), and the Real-World Product (RWP) vs. Prototype Table.

2. Overall Description

2.1 Product Perspective

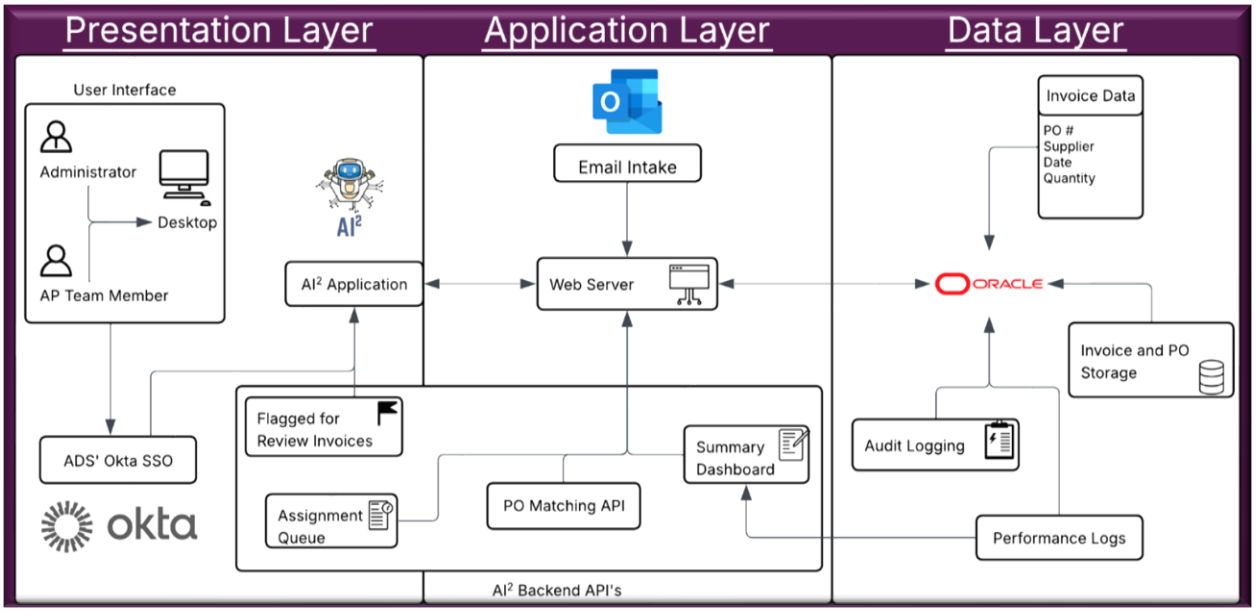
AI^2 shall be a web-based monolithic application. In the presentation layer, the user interface shall consist of dashboards that correspond to the different user roles. This shall keep access for each user limited to what is within their own purview. Verification would be performed via Okta SSO. The dashboards shall retrieve information to share with the users after it has been processed in the application layer.

The application layer shall consist of the web server, invoice parsing, the PO matching API, email intake, and the assignment queue. The email intake would be performed by integrating with Outlook's Graph API. The PO matching API, as the name implies, shall handle both the AI and hard-coded matching rules. The assignment queue shall manage assigning matches to the individual AP team members for further review. The information that is fed through the application layer shall come from the data layer.

The data layer shall consist of the database with the invoice and PO storage, as well as the audit and performance logs. The incoming information from the emails shall be parsed using an AI model, formatted into JSON files, and then broken up into structured columns to be input into the database tables. These different layers are represented in the following figure.

Figure 1

Major Functional Component Diagram



2.2 Product Functions

AI²'s primary features shall pertain to the invoice parsing and matching processes. The following tables break down each feature and express whether it will continue to be employed.

Table 1

Section 1 of RWP vs Prototype Table

RWP vs Prototype

Feature	Real World Product	Prototype	Notes
Outlook Email Integration (Graph API)	Fully Functional	Partially Functional	Simulate Intake via PDFs and local folder ingestion
Invoice Parser and Data Extractor	Fully Functional	Fully Functional	Show Progress On Various Formats over the course of the semester
Automated PO Matching Algorithm	Fully Functional	Fully Functional	
Invoice Quality Analyzer	Fully Functional	Fully Functional	
Queue Assigning and Routing	Fully Functional	Partially Functional	Start with a simple container, build from there
Queue Assignment Dashboard	Fully Functional	Partially Functional	Time constraint, Build a Minimal Viable Dashboard

As per the above table, the Graph API and Queue Assignment and Routing shall be partially implemented. The Invoice Parser and Data Extractor, the Automated PO Matching Algorithm, and the Invoice Quality Analyzer shall be fully implemented.

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Table 2

Section 2 of RWP vs Prototype Table

RWP vs Prototype

Feature	Real World Product	Prototype	Notes
Model Training and Deployment Pipeline	Fully Functional	Partially Functional	Training Workflow Demonstrated
Model Version Control	Fully Functional	Fully Functional	
AI Performance Dashboard	Fully Functional	Partially Functional	Core Metrics Dashboard
Invoice Quality Analyzer	Fully Functional	Fully Functional	
Invoice Data Anonymization Pipeline	Fully Functional	Eliminated	Limited access

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The Model Training and Deployment Pipeline, as well as the AI Performance Dashboard shall be partially implemented. The Model Training and Deployment Pipeline will ensure that a training workflow will be demonstrated, and the AI Performance Dashboard shall only display core metrics. The Invoice Data Anonymization Pipeline shall be eliminated. The Model Version Control and the Invoice Quality Analyzer shall be fully implemented.

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Table 3

Section 3 of RWP vs Prototype Table

RWP vs Prototype

Feature	Real World Product	Prototype	Notes
Admin Configuration Dashboard	Fully Functional	Partially Functional	Confidence Threshold, Queue routing
System Health and Performance Dashboard	Fully Functional	Partially Functional	Core elements like, error count, matching success, invoice processed, etc.
Error Logging and Recovery Module	Fully Functional	Fully Functional	
Admin System Alert Notifications	Fully Functional	Partially Functional	Push notifications

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The Administrator Configuration Dashboard, System Health and Performance Dashboard, and Administrator System Alert Notifications shall be partially implemented. The Administrator Configuration Dashboard shall only allow the Administrator to adjust the confidence thresholds and the queue routing with this partial implementation. The System Health and Performance Dashboard shall include core considerations, such as number of invoices processed, and matching success with this partial implementation. The Administrator System Alert Notifications shall only include push notifications. The Error Logging and Recovery Module shall be fully implemented.

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Table 4

Section 4 of RWP vs Prototype Table

RWP vs Prototype

Feature	Real World Product	Prototype	Notes
Automated Supplier Notifications	Fully Functional	Eliminated	No external comms
Automated AP Team Reminders	Fully Functional	Partially Functional	Push notification
AI Confidence Scoring Module	Fully Functional	Fully Functional	
Continuous Learning From Human Interaction	Fully Functional	Fully Functional	
Human Review Panel	Fully Functional	Fully Functional	

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The Automated AP Team Reminders shall be partially implemented, in that the AP Team Reminders shall be sent as push notifications only. The Automated Supplier Notifications shall be eliminated. The AI Confidence Scoring Module, Continuous Learning from Human Interaction, and Human Review Panel shall be fully implemented.

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Table 5

Section 5 of RWP vs Prototype Table

RWP vs Prototype

Feature	Real World Product	Prototype	Notes
Encryption Verification Tool	Partially Functional	Eliminated, Out of Scope	Most encryption handled by external systems (AWS)
AWS GovCloud Integration Tool	Fully Functional	Eliminated	Outside the scope of one semester
Audit Log Management System	Fully Functional	Partially Functional	Simulated, Basic logic to DB or file
Compliance Changelog System	Fully Functional	Partially Functional	Changes tracked manually
Auto Generated Compliance Report	Fully Functional	Eliminated	Mocked output, sample JSON
Compliance Regulations Monitoring Tool	Fully Functional	Eliminated	Out of Scope

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The Audit Log Management System and Compliance Changelog Systems shall be partially implemented. The Audit Log Management System shall be simulated. The Compliance Changelog System shall just be tracked manually rather than be its own automated process. The Encryption Verification Tool, AWS GovCloud Integration Tool, Automated Generated Compliance Reporting, and Compliance Regulations Monitoring Tool shall be eliminated.

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Table 6

Section 6 of RWP vs Prototype Table

RWP vs Prototype

Feature	Real World Product	Prototype	Notes
Okta SSO Authentication Module	Fully Functional	Eliminated	Specific to ADS, Would require their authorization
Alteryx-Oracle Integration Module	Fully Functional	Eliminated	Specific to ADS, Would require their authorization
Integration Monitoring Dashboard	Fully Functional	Eliminated	Specific to ADS, Would require their authorization

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The Okta SSO Authentication Module, Alteryx-Oracle Integration Module, and Integration Monitoring Dashboard shall all be eliminated.

2.3 User Characteristics

AI² shall have four primary user roles. There shall be administrators, accounts payable team members, AI model trainers, and compliance officers. Each user role shall have a corresponding dashboard that they shall be able to access via a login. The administrators shall have the ability to configure AI² settings to align with company goals as well as the ability to view and manage invoice matching queues and system health. The administrator dashboard shall only partially implement these. The administrators should have, at minimum, moderate technical expertise. The accounts payable team members shall have access to view and further process invoice matches that have been assigned to them. The accounts payable team members shall not

have access to other user roles' dashboards. The accounts payable team members shall, at the very least, have basic technical expertise.

The AI model trainer shall have access to adjust training parameters and to monitor AI metrics to ensure that they align with company goals. The dashboard shall only reflect the metrics. The AI model trainer shall have moderate to high technical expertise. The compliance officer shall ensure the system is adhering to federal, state, and company rules and standards. The compliance officer dashboard shall not be implemented.

2.4 Constraints

Constraints to the development of AI^2 primarily pertain to time and availability of access. Lack of access to systems that AI^2 would otherwise be integrated with (such as Okta SSO, Graph API, etc.) prevents further development of that nature. It is likely that if further explored, there would be difficulty to find around integrating with some of these systems. The tight timeframe to progress presents limits on building the prototype, though likely would present differently if applied to development of the real-world product.

2.5 Assumptions and Dependencies

AI^2 uses Flask, SQLAlchemy, and SQLite. It also uses TailwindCSS with Jinja2 in the frontend. For the AI models, it utilizes Liquid AI's with LFM2.5-VL-1.6B-Q8_0 and mmproj-LFM2.5-VL-1.6b-Q8_0 and llama.cpp. There are additional dependencies to produce dummy data, but they shall not transfer to the real-world product.